

Cloud + DevOps Master Roadmap

University of Peradeniya · Computer Engineering · 3rd Year

Commitment: 50 hrs/week · Duration: 9 months · Goal: Remote USD DevOps/Cloud role
(secondary: WSO2, Sysco LABS, IFS, Virtusa)

Current status

Phase 1 — Foundations (in progress)

■ **Linux fundamentals** — Linux Journey, OverTheWire Bandit 0–20

■ **Bash scripting** — shellscrip.sh curriculum complete

■ **Next task:** addressbook exercise (colon-delimited flat file)

Then: backup script (Bash) → system monitor (Python) → CS50P

Core principles

These principles are pulled from what actually works — verified across hiring-manager feedback and the two source videos. If any phase conflicts with them, the principle wins.

1. One growing hero project, not six tutorial projects

Do **not** learn tools in isolation. Every new phase must extend the previous phase's work. The backup script from Phase 1 grows into a Lambda-backed service in Phase 2, gets containerised in Phase 3, gets a CI/CD pipeline in Phase 4, gets provisioned by Terraform in Phase 5, and gets monitored in Phase 6. By month 9, you have **one portfolio project** that touches every skill — not six disconnected toys. This is the single biggest lesson from Video 2.

2. Build first, then certify

Certifications validate skill; they do not create it. Never study for an exam without also building the thing. Every phase project should be done **before** attempting the phase's exam. This inverts the usual cert-chaser mindset and produces engineers who can actually work.

3. 5 services, not 500

AWS has 200+ services. You only need ~5 to be job-ready as a junior: **EC2, S3, VPC, IAM, and one of {Lambda, RDS}**. Ignore the noise. Master these deeply — networking, permissions, cost — before adding anything else.

4. Concepts over syntax

Terraform syntax will change. HCL v2 → v3 → v4. Docker Compose v1 → v2. The **concepts** (declarative infrastructure, idempotency, immutable containers, orchestration) do not change. Prioritise the 'why' — the 'how' can be re-learned in a weekend.

5. Public work beats private brilliance

A public GitHub with commits every week and a blog post per phase beats a perfect private portfolio, every time. Hiring managers Google you. Give them something to find.

The Hero Project Chain

This is the concrete application of Principle 1. One idea, extended across every phase, deployed by month 9. Suggested project: **a self-hosted backup + monitoring service**.

Phase	What you add	Result at the end of the phase
1 — Foundations	Bash script that backs up a folder, timestamps, logs, rotates old files.	Runs on your Ubuntu laptop via cron. Public on GitHub.
2 — AWS Core	Rewrite in Python. Deploy as a Lambda function. Store backups in S3 with lifecycle rules. IAM role, VPC endpoint, CloudWatch logs.	Serverless, running on AWS free tier. Costs ~\$0/month.
3 — Containers	Add a small companion API (Flask/FastAPI) that lists and restores backups. Containerise both. Docker Compose locally, Kubernetes on k3s.	Multi-container app deployed to your k3s cluster on Oracle Cloud free tier.
4 — CI/CD	GitHub Actions pipeline: on push → run tests → build Docker image → push to Docker Hub → deploy to k3s. Add Trivy scan.	Push code, watch it deploy. Zero manual steps. Security-scanned.
5 — IaC	Terraform provisions the AWS infra (S3 bucket, Lambda, IAM, VPC). Ansible configures the k3s node. Modules, remote state.	<code>`terraform destroy`</code> and <code>`terraform apply`</code> re-creates the entire environment in <10 min.
6 — Observability	Prometheus + Grafana on the k3s cluster. Dashboards for backup success rate, Lambda cold starts, API p95 latency. Alertmanager to email.	Full production-grade project. README with architecture diagram. This goes on your CV.

Phase 1 — Foundations

Months 1–2 · Linux · Bash · Git · Python · YAML · Networking

Everything downstream sits on this foundation. Do not skip. Your CE degree covers networking basics — compress that sub-topic. You've already finished Linux and Bash — the addressbook exercise is your next task.

What to learn

- **Linux fundamentals** — filesystem, permissions, processes, systemd, log grepping, debugging under pressure (done)
- **Bash scripting** — variables, arithmetic, loops, expansion order, error handling (done)
- **Git & GitHub** — branches, PRs, rebasing basics, commit hygiene, SSH auth
- **Python** — CS50P syllabus (DevOps libs like boto3/argparse deferred to Phase 2)
- **YAML & JSON** — indentation, anchors, lists vs. maps. Used everywhere later.
- **Networking** — TCP/IP, DNS, HTTP/S, ports, CIDR. Your CE degree already covers most of this.

Where to learn it (free-first)

Resource	Type	Cost
Linux Journey	Free tutorial	Free
OverTheWire Bandit	Free hands-on	Free
shellscript.sh	Free tutorial	Free
KodeKloud Linux Basics	Free course + badge	Free
Harvard CS50P (Python)	Free course + cert	Free
The Missing Semester (MIT)	Free video course	Free
Learn Git Branching	Free interactive	Free
GitHub Foundations exam	Cert (waived for students)	\$99 → Free with GitHub Student Pack

Projects (build in order)

- **Addressbook exercise** — colon-delimited flat file. Practices everything from shellscript.sh. ← next task
- **Backup script (Bash)** — the seed of the hero project. Timestamps, log rotation, error handling.
- **System monitor (Python)** — CPU/RAM watcher, email on threshold. Runs as a systemd service.
- **Dotfiles repo** — your ``.bashrc``, ``.vimrc``, tmux config, tracked in Git. Real DevOps engineers publish these.

Milestone check

You can write a Bash script that safely handles filenames with spaces, use read loops without while IFS= traps, and explain what happens when you type `ls | grep foo` at a process/file-descriptor level. Also: you have a public GitHub with at least 3 commits per week for the last month.

Phase 2 — AWS Cloud Core

Months 3–4 · 5 core services, deeply — not 500 services shallowly

Skip the certification-first trap. Build first, cert to prove it. Focus on EC2, S3, VPC, IAM, plus Lambda. Everything else is variations on these primitives.

What to learn

- **EC2** — instance types, key pairs, security groups, EBS, AMIs
- **S3** — buckets, lifecycle rules, versioning, presigned URLs, static hosting
- **VPC** — subnets, route tables, NAT, security groups vs. NACLs, VPC endpoints
- **IAM** — users, roles, policies, least privilege, assume-role, MFA
- **Lambda + API Gateway** — event triggers, IAM roles, CloudWatch logs
- **CloudWatch** — metrics, log groups, alarms
- **Python for AWS** — boto3, argparse, os, subprocess, json, yaml, requests

Where to learn it (free-first)

Resource	Type	Cost
AWS Skill Builder (free tier)	Free official courses	Free
AWS Educate	Free labs + \$100 credits + 50% voucher	Free (need student email)
AWS Cloud Quest — Cloud Practitioner	Free gamified labs	Free
Andrew Brown — CLF-C02 (freeCodeCamp)	Free 14-hr video course	Free
Stephane Maarek — Ultimate AWS CLF-C02	Paid Udemy (often on sale)	~\$15 on sale
AWS Free Tier	12 months of free real usage	Free

Projects (extending the hero project)

- **Static portfolio on S3 + CloudFront** — Route 53 custom domain. Standalone warmup project.
- **Backup service on Lambda + S3** — ← **hero project extension**. Rewrite the Bash backup script as a Python Lambda. Trigger from EventBridge cron. Store in S3 with lifecycle rules.
- **Serverless URL shortener** — Lambda + API Gateway + DynamoDB. Optional but a strong CV entry.

Certification (optional but recommended for a career-changer)

AWS Certified Cloud Practitioner (CLF-C02)

- Cost: **\$100 USD** (~Rs. 32,800). With 50% AWS Educate student voucher: **\$50** (~Rs. 16,400).

- Format: 65 questions, 90 min, 700/1000 to pass
- Perk: passing unlocks a **50% off voucher** for your next AWS exam.
- Note: the 100%-free AWS Educate voucher program **expired 31 Aug 2025**. 50% is now the best publicly-available discount.

Phase 3 — Containers & Kubernetes

Months 5 · Docker → Kubernetes basics

Non-negotiable for any DevOps job. Master 20% of Docker/K8s that covers 80% of real use cases — do not chase deep K8s internals yet (CKA-level depth can wait until you have a job).

What to learn

- **Docker** — images, layers, Dockerfile, ENTRYPOINT vs CMD, volumes, networks
- **Docker Compose** — multi-container apps, healthchecks, depends_on
- **Kubernetes core** — Pods, Deployments, Services, ConfigMaps, Secrets, Ingress
- **kubectl fluency** — you should be able to write manifests without Googling basic syntax
- **Registries** — Docker Hub, GitHub Container Registry, AWS ECR

Where to learn it (free-first)

Resource	Type	Cost
Docker docs — official getting-started	Free	Free
KodeKloud — Docker for Beginners	Free course + badge	Free
KodeKloud — Kubernetes for Beginners	Free course + badge	Free
Play with Docker	Free browser labs	Free
Killercodea	Free K8s labs in-browser	Free
Kubernetes.io tutorials	Free official docs	Free
TechWorld with Nana — K8s crash course	Free YouTube	Free

Projects (extending the hero project)

- **Dockerise Phase 2 code** — ← [hero](#). Write a Dockerfile for the backup Lambda's local dev version. Add a Flask/FastAPI companion that lists and restores backups. Docker Compose runs both + Postgres.
- **Deploy on Minikube** — write Deployment, Service, Ingress manifests. Get it running locally.
- **Migrate to k3s on Oracle Cloud free tier** — ← [hero](#). Real cluster, real public URL.

Skip these (for now)

CKA / CKAD / CKS exams (~\$395 each). Also KCNA (~\$250). Get the free KodeKloud badges instead. Real Kubernetes certs are worth it once you have production experience — do not spend money on them as a junior. Your public k3s home lab beats a KCNA badge for hiring signal.

Phase 4 — CI/CD Pipelines

Months 6 · GitHub Actions primary, Jenkins as backup

The heart of DevOps. But be careful — do not fall into the trap of learning every CI/CD tool. Pick **GitHub Actions** and go deep. Learn Jenkins only if the specific role you want requires it.

What to learn

- **GitHub Actions** — workflows, jobs, matrix builds, secrets, environments, reusable workflows, OIDC to AWS
- **Pipeline patterns** — build → test → scan → package → deploy
- **Deployment strategies** — blue/green, canary, rolling; rollbacks
- **Security scanning** — Trivy (containers), Checkov (IaC), Dependabot
- **Optional:** Jenkins (only if a target role requires it), GitLab CI (similar to GH Actions)

Where to learn it (free-first)

Resource	Type	Cost
GitHub Actions official docs + Learning Lab	Free	Free
GitHub Actions — freeCodeCamp full course	Free YouTube	Free
KodeKloud — Jenkins	Free course + badge	Free
CloudBees University — Jenkins	Free cert	Free
Trivy docs	Free	Free
Checkov docs	Free	Free

Projects (extending the hero project)

- **End-to-end GitHub Actions pipeline** — ← **hero**. Push code → run pytest → build Docker image → Trivy scan → push to registry → SSH/kubectl deploy to k3s. This is the top-tier portfolio artefact of Phase 4.
- **Add signed commits + branch protection** — required for real work; visible on your GitHub profile.
- **OIDC to AWS from Actions** — no long-lived AWS keys. This alone impresses interviewers.

Phase 5 — Infrastructure as Code

Months 7 · Terraform + Ansible

Stop clicking. Everything you built in Phases 2–4 gets re-provisioned from code. This is where you become genuinely dangerous.

What to learn

- **Terraform** — providers, resources, state (remote backend on S3 + DynamoDB lock), modules, workspaces, data sources, for_each/count
- **Terraform patterns** — module structure, versioning, plan/apply workflow in CI
- **Ansible** — inventory, playbooks, roles, handlers, Jinja2 templating, idempotency
- **Concepts** — declarative vs imperative, drift, immutable vs mutable infrastructure
- **Aware of:** OpenTofu (Terraform's open-source fork after HashiCorp's BSL license change). Not required, just know it exists.

Where to learn it (free-first)

Resource	Type	Cost
HashiCorp Learn — Terraform tutorials	Free official	Free
KodeKloud — Terraform for Beginners	Free course + badge	Free
KodeKloud — Ansible for Beginners	Free course + badge	Free
Ansible official docs	Free	Free
Stephane Maarek — Terraform Associate (Udemy)	Paid (often on sale)	~\$15 on sale

Projects (extending the hero project)

- **Terraform all the AWS infra** — ← **hero**. S3 buckets, Lambda, IAM, VPC — everything from Phases 2–3 now provisioned by terraform apply. Destroy and recreate with one command.
- **Ansible playbook for the k3s node** — installs Docker/k3s, hardens SSH, sets up firewall, deploys base config.
- **Combine all three** — Terraform provisions → Ansible configures → GitHub Actions deploys. This is the phrase 'IaC + config mgmt + CI/CD' on your CV, backed by code.

Certification (worth it — high ROI)

HashiCorp Certified: Terraform Associate (003 / 004)

- Cost: **\$70.50 USD** (~Rs. 23,100). One of the highest ROI certs in the market.
- Format: ~57 questions, 60 min, online proctored via PSI
- Validity: 2 years
- Widely referenced in DevOps job postings — this one's worth paying for.

Phase 6 — Observability, Security & Advanced AWS

Months 8–9 · Prometheus · Grafana · IAM depth · SAA prep

This is what separates juniors from mid-level. You do not need to be an SRE, but you need to understand how production systems are watched, secured, and debugged.

What to learn

- **Prometheus** — scrape configs, PromQL basics, exporters, alerting rules
- **Grafana** — dashboards, data sources, variables, alerting
- **Logging** — CloudWatch Logs, log aggregation basics, structured logging
- **Cloud security** — IAM policy JSON, Secrets Manager, KMS basics, VPC endpoints
- **Advanced AWS** — EKS overview, ECR, Route 53 routing policies, CloudFormation vs Terraform trade-offs
- **Cost awareness** — Cost Explorer, right-sizing, S3 lifecycle, Reserved Instances basics

Where to learn it (free-first)

Resource	Type	Cost
Prometheus official docs	Free	Free
Grafana Academy — free badges	Free courses + cert badges	Free
PromLabs — PromQL cheat sheet	Free	Free
AWS Skill Builder — Security path	Free official	Free
Stephane Maarek — SAA-C03 (Udemy)	Paid (frequently on sale)	~\$15 on sale
Adrian Cantrill — SAA-C03	Paid deep-dive	~\$40

Projects (the finale)

- **Full observability on the hero project** — ← **hero final**. Prometheus + Grafana on k3s. Dashboards for backup success rate, Lambda cold-starts, API latency p50/p95/p99. Alertmanager → email.
- **README with architecture diagram** — draw.io or Excalidraw. This is what recruiters actually open first.
- **Blog post: 'What I built in 9 months and what broke'** — post to dev.to, LinkedIn, Hashnode. Interview magnet.
- **Cloud Cost Optimisation write-up** — analyse your own hero-project's AWS bill (or a simulated one). Right-size, add S3 lifecycle rules, document savings. Shows business maturity.

Certification (the flagship)

AWS Certified Solutions Architect – Associate (SAA-C03)

- Cost: **\$150 USD** (~Rs. 49,200). With the 50% voucher from passing Cloud Practitioner: **\$75 (~Rs. 24,600)**.
- Format: 65 questions, 130 min, 720/1000 to pass
- Validity: 3 years
- Combined with the Terraform Associate, this makes you genuinely competitive globally.

Certification timeline & 2026 costs

Prices verified via official sources in September 2026. USD → LKR at 328 (approximate mid-market rate).

Month	Certification	List Price (USD)	With discounts	Approx. LKR
1–2	GitHub Foundations (GH-900)	\$99	Free — GitHub Student Pack	Free
1–2	Harvard CS50P certificate	Free	Free	Free
1–2	KodeKloud Linux + Bash badges	Free	Free	Free
3–4	AWS Certified Cloud Practitioner (CLF-C02)	\$100	\$50 with 50% AWS Educate voucher	Rs. 16,400
5	KodeKloud Docker + Kubernetes badges	Free	Free	Free
6	CloudBees Jenkins certificate	Free	Free	Free
7	HashiCorp Terraform Associate (003/004)	\$70.50	No student discount available	Rs. 23,100
8–9	AWS Solutions Architect Associate (SAA-C03)	\$150	\$75 with 50%-off retake voucher	Rs. 24,600

Realistic total (with all discounts applied): ~Rs. 64,100 (vs. Rs. 43,500 in the older roadmap — the increase is because AWS Educate's 100%-free voucher program ended 31 Aug 2025, and the student rate is now 50%-off rather than free).

Money-saving playbook

1. Apply to the **GitHub Student Developer Pack** immediately if you haven't — free GH Foundations voucher plus dozens of tools.
2. Sign up for **AWS Educate** with your university email — \$100 credits, free labs, and the 50% exam voucher.
3. Buy Stephane Maarek's Udemy courses only during flash sales (~\$15). Never at list price (\$100+).
4. Watch AWS re:Invent and AWS Summit for occasional 50% voucher drops.
5. Pass CLF-C02 first — the 50%-off voucher for the next exam is automatic.

Job hunt (Month 8+)

Where to apply

Sri Lanka (local, in-person or hybrid)

- **WSO2** — strongest brand, serious cloud/open-source work
- **Sysco LABS** — best engineering culture, real AWS-scale problems
- **IFS** — stable, enterprise experience
- **Virtusa** — easier entry point; move on after 1–2 years
- **hSenid, 99x, Creative Software** — mid-size, look at devjobs.lk
- Job boards: **devjobs.lk**, LinkedIn Jobs (filter: Cloud/DevOps/Sri Lanka)

Remote USD roles

- [Turing](#) — vetted remote roles, US clients
- [We Work Remotely](#) — high-quality remote listings
- [Upwork](#) — freelance DevOps, start early to build reviews
- [RemoteOK](#), [Working Nomads](#), [Himalayas](#) — DevOps filters
- LinkedIn 'Remote' filter with 'DevOps' or 'SRE' keywords

Interview practice

- **Take interviews before you feel ready.** The first 5 will be rough. That's the point.
- **Practise on Pramp / interviewing.io** — free peer mock interviews.
- **Have a story for every project** — 'what went wrong', not just 'what I built'.
- **Know your salary floor** — willing to accept 30% less for your first role is realistic. Double it after 12 months.

Public presence (do this in parallel from month 1)

- **GitHub** — commits every week; the hero project public from day 1.
- **Blog** — one post per phase. Use dev.to or hashnode.com. Free, indexes in Google.
- **LinkedIn** — post every cert the day you earn it. Post project milestones. Twice a week minimum.
- **X/Twitter (optional)** — join the DevOps and cloud-native tags; don't force it if not your style.

Weekly time split (50 hrs/week)

Activity	Hours/week	Notes
Structured learning (courses, docs)	18	Down from 20 — some phases you already know.
Hands-on terminal / labs	15	The real learning happens here.

Activity	Hours/week	Notes
Hero project work	12	Up from 10 — this is the differentiator.
Writing (blog + LinkedIn)	3	Non-negotiable. This is your marketing.
Review + plan next week	2	Sunday evening ritual.

This roadmap is a living document. Update the checkboxes and add new learnings after each phase. When something changes in the industry (pricing, exam versions, tool licences), update this file rather than trusting memory.

Sources: Naxdun's previous roadmap files, insights from 2 YouTube roadmaps, and current pricing verified via web search in September 2026.